

The Okounkov-Vershik approach to the representation theory of the symmetric groups

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Introduction

This write-up is based on the work of A.M. Vershik and A.Yu. Okounkov on a new approach to the representation theory of the symmetric groups. The advantage of this approach over the conventional method of finding the irreducible representations of the symmetric groups is that it takes into account the fact that the symmetric groups form an inductive chain so that the theory could also be built inductively. We construct the irreducible representations, using the concepts of Gelfand-Tsetlin algebras, the Gelfand-Tsetlin basis, and Young-Jucys-Murphy elements. In this approach, the Young diagrams and standard Young tableaux come into the theory naturally because the branching graph for the symmetric groups and the Young graph of Young diagrams coincide. As a consequence of this, we observe that the bases for irreducible complex finite dimensional representations of the symmetric groups are indexed by standard Young tableaux.

Branching Graph and Gelfand-Tsetlin algebra

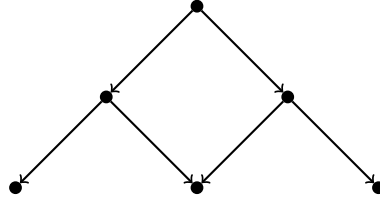
In this section we study the branching graph of irreducible representations of symmetric groups and introduce the building blocks *GZ-basis* for the irreducible representations of symmetric groups. We know that the symmetric groups S_n form an inductive chain of groups;

$$\{1\} = S_1 \subset S_2 \subset \cdots \subset S_n \subset \cdots$$

We define the *branching graph* $\Gamma = (V, E)$ of the above chain as follows. Let $V = \bigsqcup_{n \geq 1} \widehat{S}_n$ where $\widehat{S}_n$ denote the set of equivalence classes of finite dimensional complex irreducible representations of S_n . Let $\lambda \in \widehat{S}_n$, $\mu \in \widehat{S}_{n-1}$. In the branching graph, μ and λ are joined by k directed edges from μ to λ if k is the multiplicity of μ in the restriction of λ to S_{n-1} is k . In fact, $k = \dim \text{Hom}_{S_{n-1}}(V^\mu, V^\lambda)$. We use the notation $\mu \nearrow \lambda$ to indicate there is an edge from μ to λ . In the next section we will see that this branching graph for the inductive chain of symmetric groups is simple, i.e., $k = 1$ or 0 .

We know that the symmetric group S_1 has only one irreducible representation: the *trivial* representation. The group S_2 has two irreducible representations: the trivial representation, and the *alternating* representation. The group S_3 has three irreducible representations: two 1-dimensional representations: the trivial and alternating representations, and one 2-dimensional representation, called the *standard representation*. The standard representation of S_3 , when restricted to S_2 , decomposes as a direct sum of the trivial and alternating representations of S_2 .

Below is the branching graph for $S_1 \subset S_2 \subset S_3$, where the node at the first level denotes the trivial irreducible representation; the two nodes at the second level denote the trivial and alternating representations of S_2 ; and the three nodes at the third level denote the trivial, standard, and alternating representations of S_3 , respectively from left to right.



Assume (for a moment) our branching graph is simple. Let V^λ be an irreducible S_n -module, where $\lambda \in \widehat{S_n}$. Then

$$V^\lambda = \bigoplus_{\mu \nearrow \lambda, \mu \in \widehat{S_{n-1}}} V^\mu.$$

By induction, we obtain a canonical decomposition

$$V^\lambda = \bigoplus_T V_T$$

indexed by all possible chains

$$T = \lambda_0 \nearrow \lambda_1 \nearrow \cdots \nearrow \lambda_n = \lambda$$

where each V_T is a S_1 module (i.e., one-dimensional subspace). If we choose a non-zero vector v_T from each V_T , we obtain a basis $\{v_T\}$ for the module V^λ . This basis is called the *Gelfand-Tsetlin basis* (GZ-basis) for V^λ .

Definition 1. Let Z_n denote the center of the group algebra $\mathbb{C}[S_n]$. Define the Gelfand-Tsetlin algebra (GZ-algebra) GZ_n to be the subalgebra of $\mathbb{C}[S_n]$ generated by $Z_1, \dots, Z_n$, i.e.,

$$GZ_n = \langle Z_1, \dots, Z_n \rangle.$$

It is clear from this definition that GZ_n is commutative.

Proposition 1. The algebra GZ_n is the algebra of all operators diagonal in the GZ-basis for all irreducible representations of S_n . In particular, it is a maximal commutative subalgebra of $\mathbb{C}[S_n]$.

A proof of the above result may be found in [1] (Proposition 1.1). We obtain the following corollary immediately from the above proposition.

Corollary 1. (i) *If u is an eigenvector of every element of GZ_n , then a scalar multiple of u belongs to the GZ -basis of V^λ .*

(ii) *If u and v are two GZ -vectors having the same eigenvalues for every element of GZ_n , then $u = v$.*

By this corollary, it is clear that GZ -vectors are determined (upto scalars) by the eigenvalues of elements of GZ_n . Note that by a GZ -vector, we mean a vector in the GZ -basis of some irreducible representation of S_n .

Young-Jucys-Murphy elements

Next, we use the following well-known proposition to show that the branching graph for the symmetric groups is simple. The proof of the following can be found in [1] (Proposition 1.4).

Proposition 2. *Let M be a semisimple finite-dimensional $\mathbb{C}$ -algebra. Let N be a subalgebra of M , and $Z(M, N)$ denote the centralizer of N in M , i.e., the subalgebra of all elements of M that commute with N . Then the following are equivalent:*

(i) *The restriction of any finite-dimensional irreducible complex representation of M to N is multiplicity-free.*

(ii) *The centralizer $Z(M, N)$ is commutative.*

Since $\mathbb{C}[S_n]$ is a semisimple algebra (if characteristic of a field k doesn't divide the order of a group G , then the group algebra $k[G]$ is semisimple), to show that the branching graph is simple, it is enough to prove that $Z(\mathbb{C}[S_n], \mathbb{C}[S_{n-1}])$ is commutative for all $n \geq 1$. For simplicity, we write

$$Z_{(n-1,1)} = Z(\mathbb{C}[S_n], \mathbb{C}[S_{n-1}]).$$

To establish the simplicity of the branching graph and to provide a detailed description of the structure of centralizers and the Gelfand–Tsetlin algebra, we introduce the Young–Jucys–Murphy (YJM) elements.

Definition 2. *For $i = 2, \dots, n$, define the YJM elements $X_i \in \mathbb{C}[S_n]$ by*

$$X_i = (1 \ i) + (2 \ i) + \dots + (i-1 \ i).$$

and $X_1 = 0$.

Observe that X_i is the difference of the sum of all transpositions in S_i and the sum of all transpositions in S_{i-1} , i.e., X_i is the difference of an element in Z_i and an element in Z_{i-1} . Therefore, $X_i \in GZ_n$. In fact, we will see that GZ_n is generated by $X_1, X_2, \dots, X_n$.

Proposition 3. (i) $Z_{(n-1,1)} = \langle Z_{n-1}, X_n \rangle$

(ii) $GZ_n = \langle X_1, X_2, \dots, X_n \rangle$.

The proof of the above result is slightly technical in nature and can be found in [4] (Theorem 4.2).

Proposition 4. *The branching graph for symmetric groups is simple.*

Proof. By Proposition 3, we have $Z_{(n-1,1)} = \langle Z_{n-1}, X_n \rangle$. Note that $X_n = T_n - T_{n-1}$ where T_n is the sum of all transpositions in S_n and T_{n-1} is the sum of all transpositions in S_{n-1} . Since $T_n \in Z_n$ and $T_{n-1} \in Z_{n-1}$, X_n commutes with every element in Z_{n-1} and consequently $Z_{(n-1,1)}$ is commutative. Then by Proposition 2, the branching graph is simple. $\square$

We already observed that the eigenvalues of elements of GZ_n determine the G -vectors up to scalars. Therefore, by the above proposition, the eigenvalues of $X_1, \dots, X_n$ determine the GZ -vectors (up to scalars). This motivates us to define the *spectrum* of S_n . The GZ -basis for an irreducible complex representation in the case of symmetric groups is sometimes called a *Young basis* and the elements of Young basis are called *Young vectors*.

Definition 3. *Let v be a Young vector. Define*

$$\alpha(v) = (a_1, \dots, a_n) \in \mathbb{C}^n,$$

where a_i is the eigenvalue of X_i on v . The vector $\alpha(v)$ is called the weight of v , and we define

$$\text{Spec}(n) := \{\alpha(v) \mid v \in \text{Young basis for some } \lambda \in \widehat{S_n}\}.$$

By Corollary 1, a Young vector v is determined (up to a scalar) by the elements of the spectrum $\text{Spec}(n)$.

Thus,

$$\dim GZ_n = |\text{Spec}(n)| = \sum_{\lambda \in \widehat{S_n}} \dim \lambda.$$

Thus we get a natural bijection between the elements of $\text{Spec}(n)$ and all paths in the branching graph of $S_1 \subset S_2 \subset \dots \subset S_n$. Let $\alpha \in \text{Spec}(n)$ be the weight vector of a Young vector v_α , and let T_α denote the corresponding path in the branching graph. Also, given a path T with the Young vector v_T , $\alpha(v_T)$ denotes its weight. Then the bijection is given by the maps:

$$\alpha \rightarrow T_\alpha \quad \text{and} \quad T \rightarrow \alpha(v_T).$$

Now we define an equivalence relation $\sim$ on $\text{Spec}(n)$ as follows:

$$\begin{aligned}\alpha \sim \beta &\iff v_\alpha \text{ and } v_\beta \text{ belong to the same irreducible } S_n\text{-module} \\ &\iff T_\alpha \text{ and } T_\beta \text{ have the same end vertex.}\end{aligned}$$

Then clearly

$$|\text{Spec}(n)/\sim| = |\widehat{S_n}|.$$

Coxeter Generators and their actions on GZ-basis

We know that S_n is generated by all transpositions. Since any transposition in S_n can be written as a product of elements of the form $(i \ i+1)$, they generate S_n . The transpositions $s_i := (i \ i+1)$ (for $1 \leq i < n$) are called *Coxeter generators*. To study the action of $\mathbb{C}[S_n]$ on Young vectors, it is enough to understand the action of the Coxeter generators of S_n on Young vectors.

It is interesting to observe that the Coxeter generators satisfy the following relations:

- (1) $s_i^2 = 1$, $s_i X_j = X_j s_i$ for $j \neq i, i+1$
- (2) $X_i X_{i+1} = X_{i+1} X_i$
- (3) $s_i X_i + 1 = X_{i+1} s_i$.

We call the above relations Coxeter relations. The relations (1) and (2) are obvious, and (3) follows because

$$\begin{aligned}s_i X_i s_i + s_i &= (i \ i+1) [(1 \ i) + (2 \ i) + \cdots + (i-1 \ i) + (i \ i+1)] (i \ i+1) + (i \ i+1) \\ &= (1 \ i+1) + (2 \ i+1) + \cdots + (i-1 \ i+1) + (i \ i+1) = X_{i+1}. \\ &\Rightarrow s_i X_i + 1 = X_{i+1} s_i.\end{aligned}$$

Since a Young vector v is characterized by $\alpha(v) \in \text{Spec}(n)$, to study the action of the Coxeter generators in v , we focus on the action of s_i upon elements of $\text{Spec}(n)$.

We will use the above Coxeter generators and the generators' relations to give explicit formulas for the actions. The following proposition describes some properties of elements of $\text{Spec}(n)$. The proof of the proposition follows almost immediately from the Coxeter relations we defined above. The interested reader may find the proof in detail in [3] (Proposition 3.3.3).

Proposition 5. *Let $\alpha = (a_1, \dots, a_n) \in \text{Spec}(n)$. Then for $i = 1, 2, \dots, n-1$:*

- (i) $a_i \neq a_{i+1}$
- (ii) $a_{i+1} = a_i \pm 1$ if and only if $s_i \cdot v_\alpha = \pm v_\alpha$

(iii) If $a_{i+1} \neq a_i \pm 1$, then $\alpha' = s_i \cdot \alpha = (a_1, \dots, a_{i-1}, a_{i+1}, a_i, \dots, a_n) \in \text{Spec}(n)$ and $\alpha \sim \alpha'$.

Also, the vector (up to scalar factor)

$$v_{\alpha'} = s_i v_{\alpha} - \frac{1}{a_{i+1} - a_i} v_{\alpha}.$$

Moreover, the space spanned by v_{α} and $v_{\alpha'}$ is invariant under the action of X_i, X_{i+1} and s_i and the matrices for them with respect to the basis $\{v_{\alpha}, v_{\alpha'}\}$ are given by

$$\begin{bmatrix} a_i & 0 \\ 0 & a_{i+1} \end{bmatrix}, \quad \begin{bmatrix} a_{i+1} & 0 \\ 0 & a_i \end{bmatrix} \text{ and } \begin{bmatrix} \frac{1}{a_{i+1}-a_i} & 1 - \frac{1}{(a_{i+1}-a_i)^2} \\ 1 & \frac{1}{a_i-a_{i+1}} \end{bmatrix}$$

respectively.

Definition 4. We say a Coxeter generator s_i is admissible for $\alpha \in \text{Spec}(n)$ if $\alpha = (a_1, \dots, a_n) \in \text{Spec}(n)$ and $a_{i+1} \neq a_i \pm 1$.

From the above proposition, it is clear that if $\alpha, \beta \in \text{Spec}(n)$ and β is obtained from α by a sequence of admissible transpositions, then $\alpha \sim \beta$.

Let H denote the algebra generated by X_{n-1}, X_n and s_{n-1} . It follows that all the irreducible representations of H is at most two dimensional. This is because if V is an H -module and v is a common eigenvector of X_{n-1} and X_n (which exists since X_{n-1} and X_n commute), then the subspace W of V spanned by v and $s_{n-1}v$ is a submodule of V (by the Coxeter relations) and is two dimensional. Note that the algebra $\mathbb{C}[S_n]$ is generated by the algebra $\mathbb{C}[S_{n-1}]$ and H . Therefore, we can carry out the study of representations of $\mathbb{C}[S_n]$ inductively from $\mathbb{C}[S_{n-1}]$ as each step from the $(n-1)^{\text{th}}$ step to the n^{th} step reduces to the study of representations of H . This reasoning, along with the proposition 5(iii) suggests that if $\alpha \in \text{Spec}(n)$ is an integral vector, then all the irreducible representations of S_n are defined over $\mathbb{Q}$ if we choose our Young basis appropriately.

Content Vectors

In this section, we relate Young tableaux to irreducible representations of S_n via content vectors.

Definition 5. Let $a = (a_1, a_2, \dots, a_n) \in \mathbb{C}^n$. We say α is a content vector of length n if it satisfies the following axioms:

- (i) $a_1 = 0$
- (ii) $\{a_i - 1, a_i + 1\} \cap \{a_1, \dots, a_{i-1}\} \neq \emptyset$ for all $i > 1$
- (iii) If $a_i = a = a_j$ with $i < j$, then $\{a_{i+1}, \dots, a_{j-1}\} \supseteq \{a - 1, a + 1\}$.

The set of all content vectors of length n is denoted by $\text{Cont}(n)$. The axiom (ii) imposes the content vector to be an integral vector, i.e., an element of $\mathbb{Z}^n$.

Example 1. We have $\text{Cont}(1) = \{0\}$ and $\text{Cont}(2) = \{(0, 1), (0, -1)\}$.

Proposition 6. For every $n \in \mathbb{N}$, $\text{Spec}(n) \subseteq \text{Cont}(n)$.

If $\alpha = (a_1, \dots, a_n) \in \text{Cont}(n)$, clearly $a_1 = 0$ as $X_1 = 0$. We can verify the conditions (ii) and (iii) by induction on n . For a detailed proof, refer [1] (Theorem 5.1).

The transposition $s_i = (i \ i + 1)$ is said to be admissible for $(a_1, \dots, a_n) \in \text{cont}(n)$ if $a_i \neq a_{i+1} \pm 1$. Like we defined an equivalence relation on $\text{Spec}(n)$, we define an equivalence relation on $\text{Cont}(n)$ as well. We say $\alpha \approx \beta$ if β can be obtained from α by a sequence of admissible transpositions.

Young graph and irreducible representations of S_n

We define the Young graph $\mathbb{Y}$ to be a graph with vertices Young diagrams and a vertex μ is joined to a vertex λ by a directed edge if $\mu \subset \lambda$ and $\lambda - \mu$ is a single box. The rows and columns of a Young diagram are numbered from top to bottom and left to right respectively. Correspondingly, each box in the Young diagram is denoted by a system of coordinates similar to the entries of a matrix.

Example 2. The following is an example of a Young diagram associated with a partition of 6 with a system of coordinates for the boxes.

(1, 1)	(1, 2)	(1, 3)
(2, 1)	(2, 2)	
(3, 1)		

Definition 6. The content of a box B with coordinates (i, j) of a Young diagram $C(B) = j - i$.

Definition 7. Let $\lambda \vdash n$ be a partition of n . A Young tableau of shape λ is an array of n integers obtained by placing $1, \dots, n$ into the boxes of the Young diagram for λ . A Young tableau with entries increasing along both rows and columns is called a standard Young tableau.

Young Diagram Lattice (till fourth level)

Proposition 7. *For any $T \in \text{Tab}(n)$, the map*

$$\phi : \text{Tab}(n) \rightarrow \text{Cont}(n)$$

defined by

$$T = \lambda_1 \nearrow \lambda_2 \nearrow \cdots \nearrow \lambda_n = \lambda \quad \mapsto \quad (C(\lambda_1), C(\lambda_2 - \lambda_1), \dots, C(\lambda_n - \lambda_{n-1})) := C(T)$$

is a bijection. Moreover, for $T_1, T_2 \in \text{Tab}(n)$,

$$T_1 \approx T_2 \iff C(T_1) \approx C(T_2).$$

This proposition follows easily as the content vector of a Standard Young tableau obviously satisfies the axioms of a content vector and given a content vector, the axioms uniquely determine the Standard Young tableau.

From the above lemma and above proposition, it follows that the $|\text{Cont}(n)/\approx|$ is the number of Young diagrams corresponding to partitions of n , which is the number of partitions of n , $p(n)$. Note that we already have $\text{Spec}(n) \subseteq \text{Cont}(n)$. If $\alpha \in \text{Spec}(n)$ and $\beta \in \text{Cont}(n)$ such that $\alpha \approx \beta$, then by the definition of $\approx$, $\beta \in \text{Spec}(n)$ and $\alpha \sim \beta$. Therefore if A and B denote two equivalence classes in $\text{Spec}(n)$ and $\text{Cont}(n)$ respectively, then either $A \cap B = \emptyset$ or $B \subseteq A$. We already observed that $|\text{Spec}(n)/\sim| = |\widehat{S_n}|$ and $|\text{Cont}(n)/\approx| = p(n)$. But the number of finite dimensional irreducible representations of S_n , $|\widehat{S_n}|$ is the number of conjugacy classes of S_n which is the number of partitions of n . Thus $|\text{Spec}(n)/\sim| = |\text{Cont}(n)/\approx| = p(n) \implies \text{Spec}(n) = \text{Cont}(n)$ and $\sim$ and $\approx$ coincide. Hence we obtain the following important proposition.

Proposition 8. *The branching graph for the chain of symmetric groups and the Young graph $\mathbb{Y}$ of Young diagrams are the same.*

Thus the Young diagrams corresponding to partitions of n represents distinct elements of $\widehat{S_n}$ and bases for all the finite dimensional complex irreducible representations of the symmetric groups are indexed by Young tableaux. Also, it is interesting to observe from the above proposition that if $\lambda \vdash n$ and $\mu \vdash (n+k)$, then the multiplicity of λ when μ restricted to S_n is the number of paths from λ to μ in $\mathbb{Y}$.

References

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